Smith & Nephew Analytics Internship Assessment

Case Study 2: SIOP Reporting Dashboard

Prepared By

Name: Sanjay D

Tool Used

Microsoft Power BI Desktop

Submission Date

June 2026

1. Introduction

Sales, Inventory and Operations Planning (SIOP) is a business process that helps organizations balance demand, inventory, and operational planning. The objective of this case study was to transform forecast and actual datasets into a meaningful reporting solution and develop an interactive Power BI dashboard that enables business users to monitor forecast performance and analyze key business metrics.

The dashboard was designed to provide visibility into Revenue Forecast, Non-Revenue Forecast, and CAPEX Forecast across different countries while enabling users to compare forecast snapshots using Resultant, Lag 0, and Lag 1 periods.

2. Objectives

The main objectives of this project were:

- Import and transform forecast and actual datasets.
- Develop a scalable Power BI reporting solution.
- Create interactive dashboards with filters and KPI indicators.
- Calculate Forecast Accuracy and Forecast Bias metrics.
- Enable users to compare forecast snapshots using lag analysis.
- Provide country-level forecast insights through visual analytics.

3. Data Preparation

Two datasets were provided:

Forecast Dataset

The Forecast dataset contains:

- Country
- Item
- AsOfPeriod
- Period
- Revenue Forecast
- Non-Revenue Forecast
- CAPEX Forecast

Actual Dataset

The Actual dataset contains:

- Country
- Item
- Period
- Actual Revenue Quantity
- Actual Non-Revenue Quantity
- Actual CAPEX Quantity

The datasets were imported into Power BI and validated using Power Query Editor. Data quality checks were performed to ensure consistency and completeness.

4. Data Modeling

Relationships were established between Forecast and Actual datasets based on common business attributes such as Country, Item, and Period.

A calculated column called Lag Type was created to classify forecast snapshots into:

- Resultant
- Lag 0
- Lag 1

This enabled comparison of forecast versions across different planning cycles.

5. KPI Development

The following key performance indicators were developed:

Forecast Accuracy %

Forecast Accuracy measures how closely forecasted values align with actual business results.

Formula:

Forecast Accuracy % = 1 - ABS(Forecast - Actual) / Actual

Forecast Bias %

Forecast Bias identifies whether forecasts tend to overestimate or underestimate actual performance.

Formula:

Forecast Bias % = (Forecast - Actual) / Actual

Total Revenue Forecast

Represents the total forecasted revenue across all selected dimensions.

Total Non-Revenue Forecast

Represents the total forecasted non-revenue values.

Total CAPEX Forecast

Represents the total forecasted capital expenditure values.

6. Dashboard Design

The dashboard was designed using business intelligence best practices and includes:

Interactive Filters

- AsOfPeriod
- Country
- Lag Type

KPI Cards

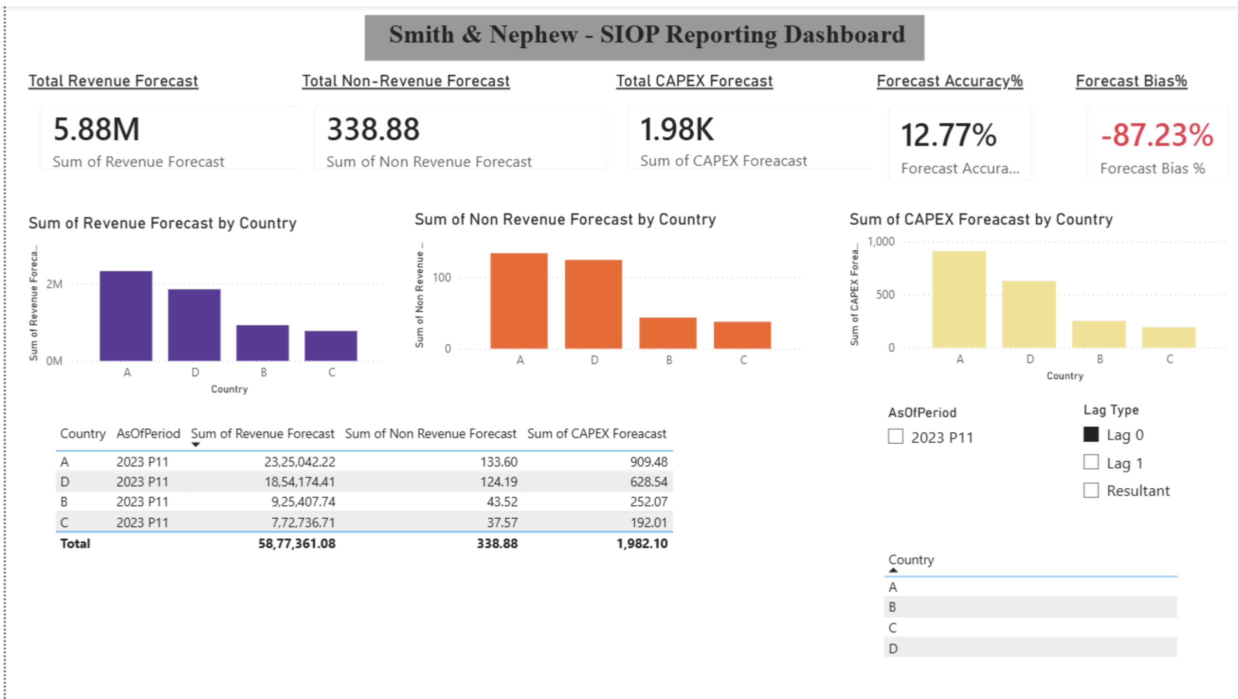
- Total Revenue Forecast
- Total Non-Revenue Forecast
- Total CAPEX Forecast
- Forecast Accuracy %
- Forecast Bias %

Visualizations

- Revenue Forecast by Country
- Non-Revenue Forecast by Country
- CAPEX Forecast by Country

Detailed Forecast Table

Provides detailed forecast information by country and planning period.



7. Analysis and Findings

Based on the dashboard analysis:

- Revenue Forecast contributes the largest share of forecast values.
- CAPEX Forecast represents a relatively smaller portion of the planning dataset.
- Forecast Accuracy indicates the degree of alignment between forecasted and actual values.
- Forecast Bias highlights areas where forecasting may require improvement.
- Country-level comparisons help identify variations in forecast performance across regions.

The implemented slicers allow business users to dynamically filter and analyze information for specific countries, periods, and lag classifications

9. Conclusion

The SIOP Reporting Dashboard successfully transforms forecast and actual datasets into an interactive analytical solution. Through the use of KPI indicators, lag analysis, and dynamic filtering, the dashboard enables users to evaluate forecasting performance and gain actionable business insights.

The solution demonstrates the practical application of Power BI, data transformation, KPI development, and business intelligence reporting techniques for real-world planning and forecasting scenarios.

Technologies Used

- Microsoft Power BI Desktop
- Power Query
- DAX (Data Analysis Expressions)
- Microsoft Excel